The genus-4 sub-rho row, re-priced in bitop/v1

`research/hec_bitop_20261020` · protocol frozen 2026-10-20 · report 2026-10-20

Question

The scoreboard's only end-to-end measurement below rho is index calculus on a genus-4 hyperelliptic Jacobian over $\mathbb{F}_{61}$ ($N = 16,790,591 \approx 2^{24}$) at 0.24× rho. It rests on two choices:

- **The in-situ unit.** Its linear algebra and smoothness oracle enter through a wall-clock calibration.
- **An 8-bit rho walk.** Round twelve later found this walk slow on the largest rows.

Two questions follow:

- Does the row stay below rho when both arms are counted where they execute and priced in one accounting convention, against the full-width walk?
- Does the genus-4 exponent hold?

`bitop/v1` is a reproducible accounting convention with frozen gate-equivalent weights. It is not a physical gate count or circuit cost; see the protocol's interpretation amendment.

Method

- **Instrument.** Every operation of both arms is counted where it executes:
  - the $\mathbb{F}_p$ additions, multiplications, squarings, inversions, square roots and Legendre symbols;
  - arithmetic modulo N;
  - hash probes and word operations.

  Both arms are priced by `src/cryptanalysis/bitops.rs`:
  - field operations at bits(p);
  - mod-N operations at bits(N).

  The work is grouped in phases: precomputation, walk, oracle, linear algebra and verification.
- **Coverage diagnostic.** Under Callgrind, the two arms' instructions per counted gate are within 1.16×. The IC oracle alone runs 1.7–2× rho's, because `fp_inv`'s exponentiation is priced as one primitive.
  - Not counted: memory movement, generator internals beyond one word operation per draw, and loop-control tests. Both arms do these.
- **Rows.**
  - The 18 rows of round seven, in the fb-walk configuration.
  - Index calculus: seed 20260916, 5 runs per row.
  - Rho: seed `0xC0FFEE`, 40 runs per row, full-width walk.
  - Every logarithm verified.
- **Analysis.** `examples/hec_bitop_analyse.rs`, with 2,000 bootstrap resamples and seed 20261020.
  - Per row: the ratio of mean costs, with a 95% interval.
  - Per genus: the slope of $\log_2\left(\frac{\text{IC}}{\text{rho}}\right)$ on $\log_2 N$, with a two-stage bootstrap interval.

Results

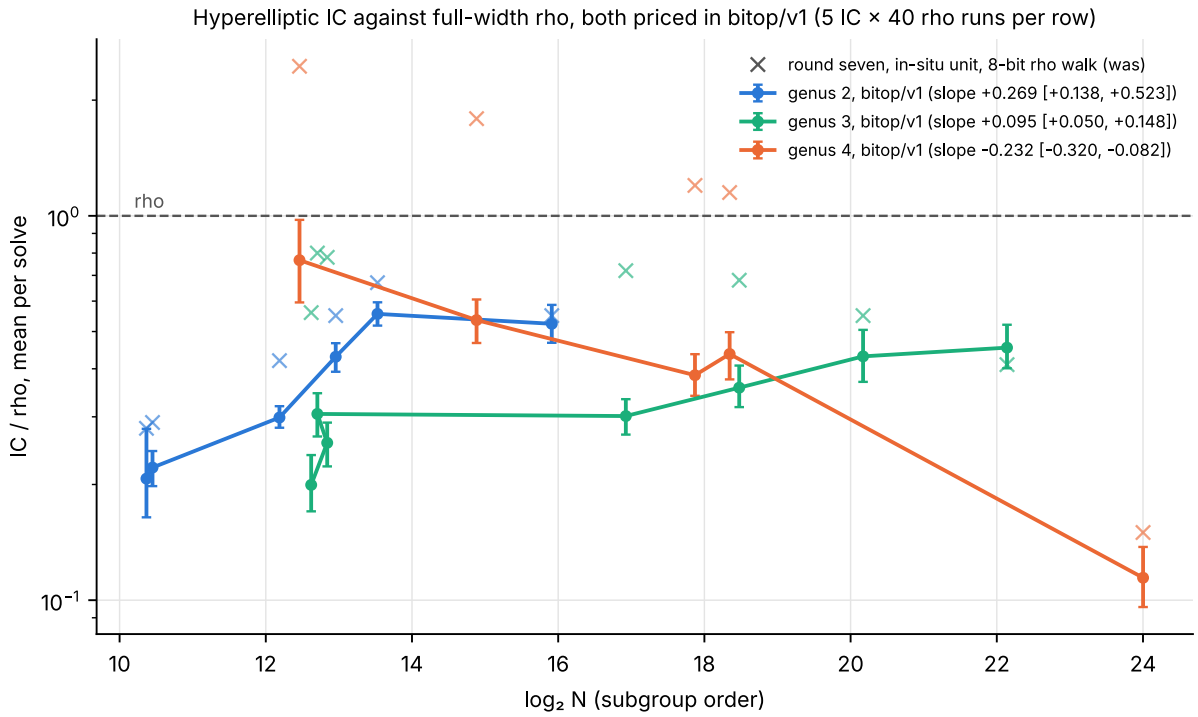


Figure 1: IC/rho per row in bitop/v1 with 95% intervals (5 IC × 40 rho runs); crosses are round seven's in-situ ratios against the 8-bit walk.

genus	bitop/v1 slope [95%]	round seven, in-situ	verdict
4 (5 rows)	−0.232 [−0.320, −0.082]	−0.29 [−0.42, −0.16]	advance holds
3 (7 rows)	+0.095 [+0.050, +0.148]	−0.01 [−0.07, +0.05]	rising, above round seven's band
2 (6 rows)	+0.269 [+0.138, +0.523]	+0.21 [0.00, +0.41]	rising

Genus 4, p = 61: IC/rho = **0.114 [0.096, 0.138]**. The upper bound is below 1, so by the preregistered rule the row **survives**.

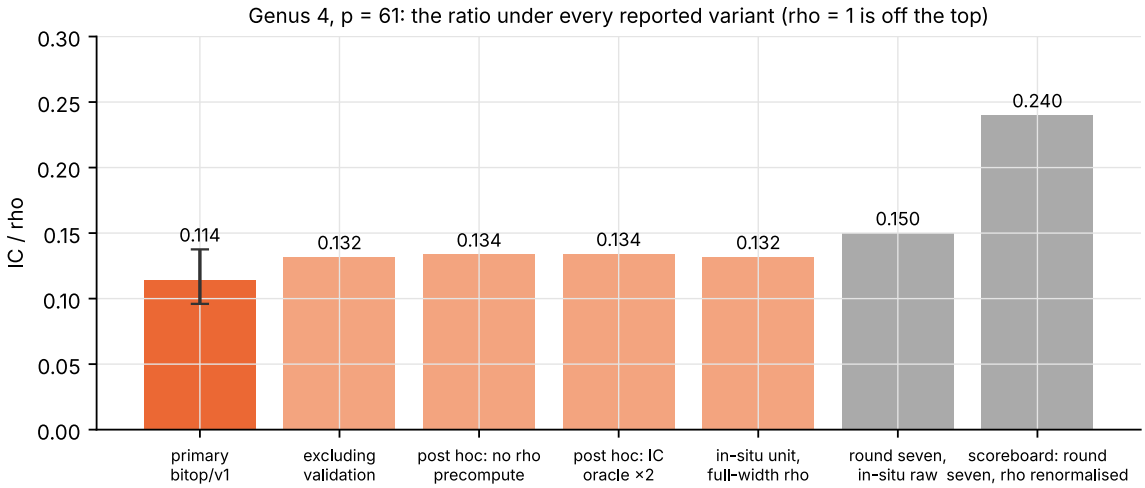


Figure 2: The p = 61 row under the primary measure and every reported variant; the grey bars are round seven's figures.

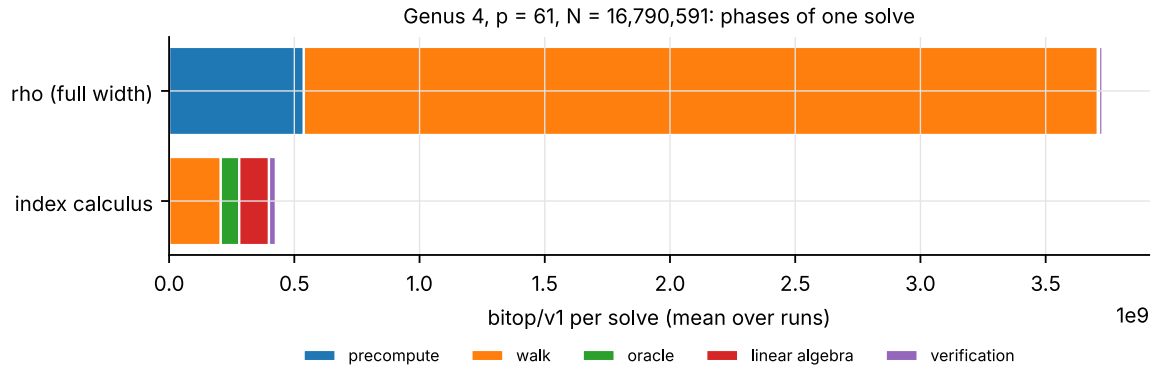


Figure 3: Where one solve's bitop/v1 goes at genus 4, p = 61 (recorded phase totals over runs, divided by runs).

What qualifies the result

- **Rho's set-up drives the small rows (post hoc).** The rho reference precomputes 16 full-width jump points.
  - At small N that fixed cost dominates: 68% of rho at genus 2, $p = 251$.
  - Without it, every genus-2 row and the small genus-3 and genus-4 rows read above 1.
  - At genus 4, $p = 61$, it is 14% of rho, and the row reads 0.134 without it.
- **The code moved.** With the in-situ unit and the full-width rho, the genus-4 rows read 1.22, 0.87, 0.61, 0.62 and 0.13. Round seven read 2.47, 1.79, 1.21, 1.16 and 0.15.
  - The move from 0.24 to 0.11 combines later algorithm changes, the corrected walk and the unit.
  - At the same head, the unit alone moves the row from 0.132 (in-situ) to 0.114 (bitop/v1).
- **Class: accounting.** No algorithm changed in this study. It confirms the row and the genus-4 exponent; it does not improve index calculus.

Scope

- Toy hyperelliptic Jacobians, with no deployed system.
- Counts only.
- One curve per row; genus 4 has five sizes, and the winning row is the largest.
- Coverage limits as stated above.
- Deviations and the owner's amendment are in `PROTOCOL.md`.